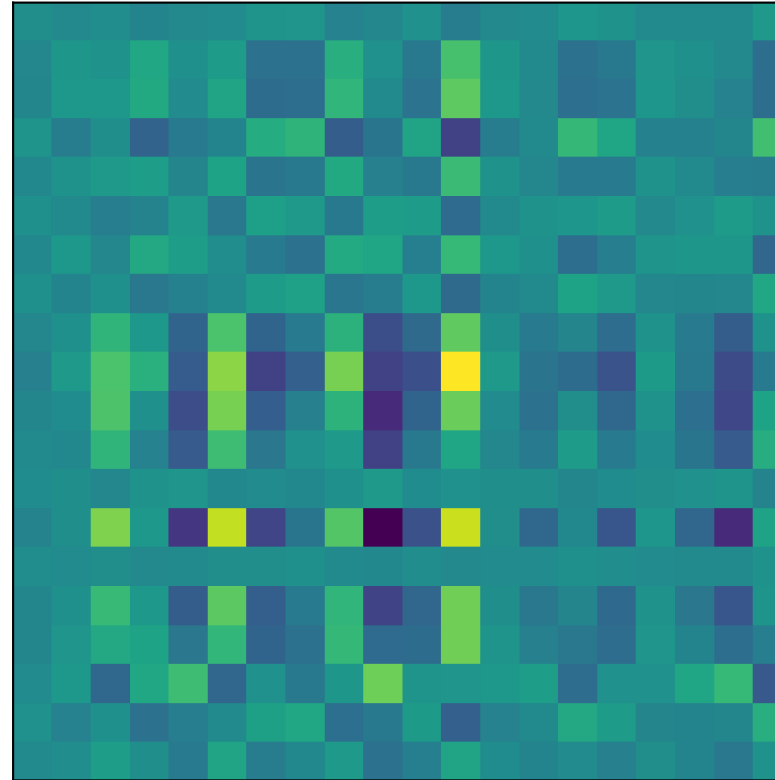
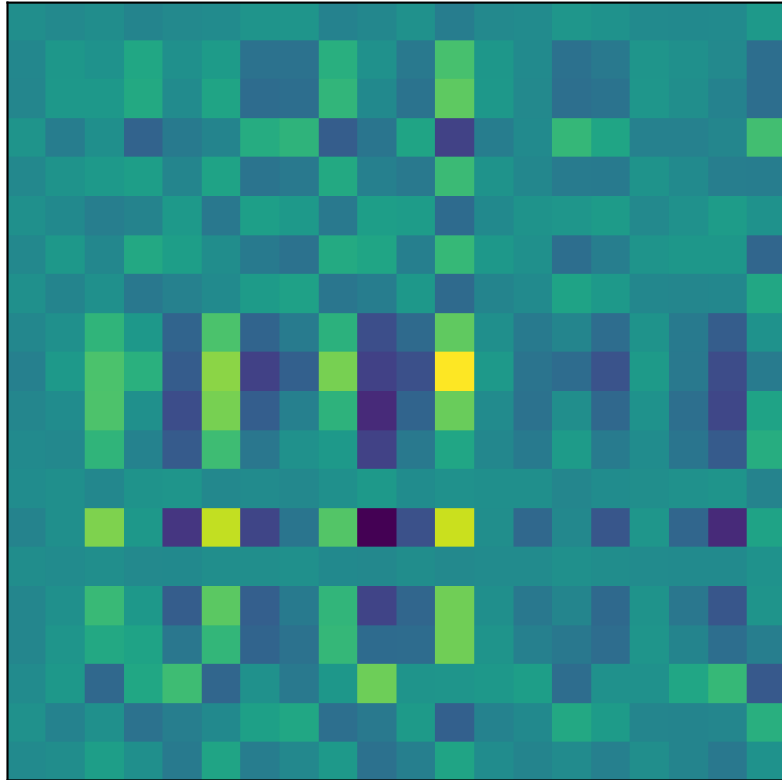
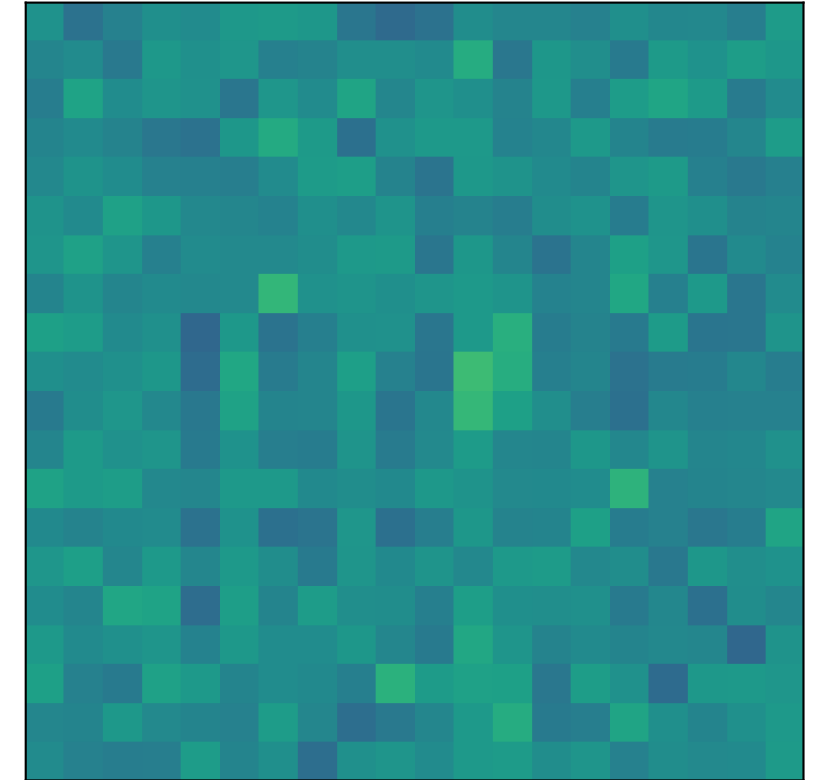


LoRA rank sweep on low-rank shifted benchmark
 $M = 20, D = M^2 = 400$, target rank=2, benchmark=rastrigin
 matched search dims: r=1: d=40, r=2: d=80, r=4: d=160
 LoRA SVD at r=2, $d = 2Mr = 80$ ($A \in \mathbb{R}^{20 \times 2}, B \in \mathbb{R}^{2 \times 20}$)
 rel. MSE=2.21e-31

Target $X_* = A_* B_*$



Best RP at r=2, $d = 80$ ($P \in \mathbb{R}^{80 \times 400}$)
 rel. MSE=7.59e-01



Setup

Full space: $D = M^2 = 400$.

At each LoRA rank r , all methods search in the same dimension $d = 2Mr$:

LoRA: rank- r factors (A, B), $d = 2Mr$
 RP: random projection P , $d = 2Mr$
 RB: random blocking groups, $d = 2Mr$

Per-rank search dimensions:

r=1 -> d=40
 r=2 -> d=80
 r=4 -> d=160

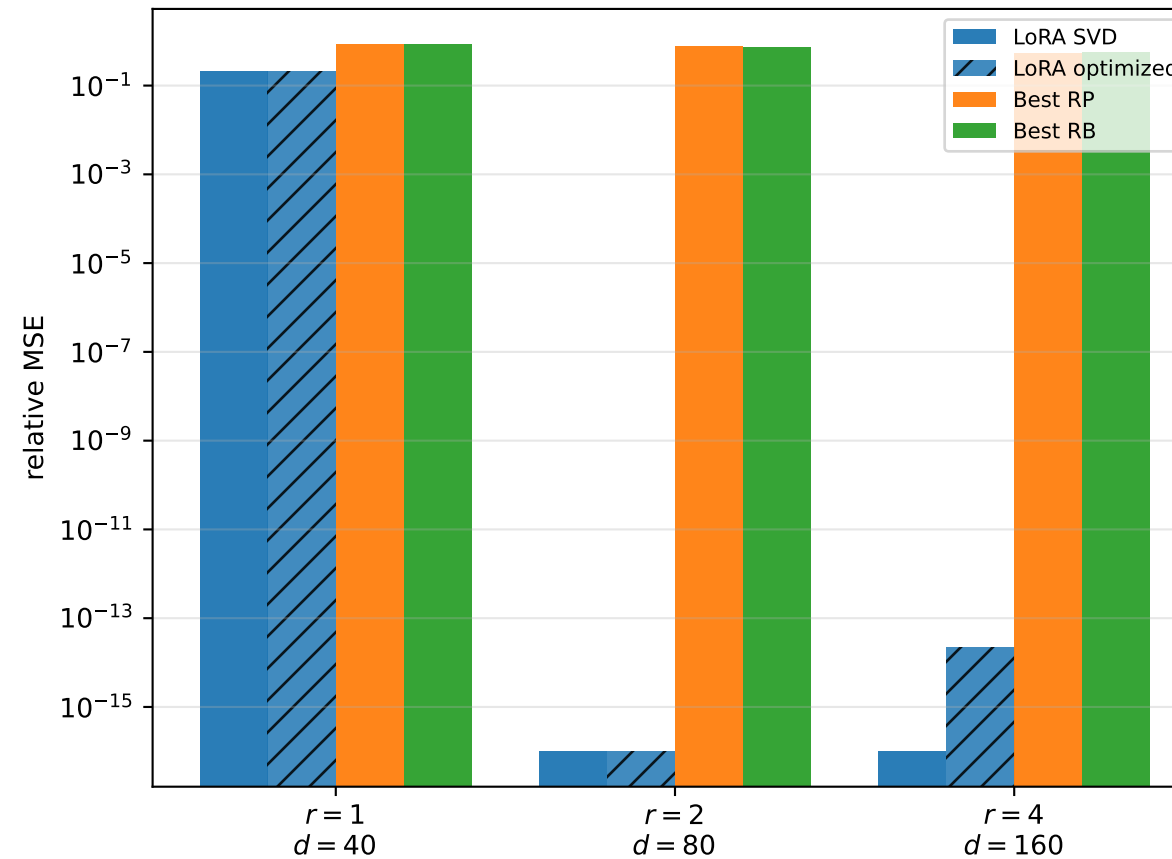
Target rank = 2.

LoRA with $r \geq$ target rank can represent the optimum exactly (SVD / factor tuning).

RP/RB use fixed random maps; at each rank they only tune z in the matched d .

At $r=2$, RB uses $d = 80$ ($D = 400 \rightarrow \{0, \dots, 79\}$).

Reconstruction error vs rank (matched d per method)



Shifted rastrigin vs rank (matched d per method)

